Evaluation of V7.4b

The diagnosis is correct, the surgery is in roughly the right place, but the spec leaves enough underspecified that it risks shipping as a half-fix.

What V7.4b gets right. The crucial intellectual move is distinguishing between *architectural failure* and *implementation failure*. V7.4 didn't fail because eigenfields are wrong — it failed because the hard mask `1_{φ_i ∈ B*_t}` is a discrete commitment in a system whose entire philosophy is continuous admissibility. That's a one-line bug in a 20-page architecture. The V7.3 evaluation document made this explicit: V7.3's strength was that β became "continuous and healthy," cash became "partial — not binary," and exposure deformed "continuously." V7.4 then collapsed back into a hard switch at exactly the wrong layer. V7.4b restores the continuous logic.

This is also philosophically aligned with MRTP. Hard Local-Star selection is the runtime equivalent of the shortest-path failure mode — it commits to a single trajectory and "destroys future recoverability." Soft top-K aggregation with softmax temperature is precisely the "recoverable corridor" generalization at the block-membership layer. The fix isn't ad-hoc; it's the MRTP principle applied one level down.

Where the mechanism is plausible. The softmax over $Q_m(t)$ with temperature τ is the right family of functions — it interpolates smoothly between uniform weighting ($\tau \to 0$) and hard selection ($\tau \to \infty$), and it integrates cleanly with the existing block-score machinery. Strategy membership $M_i(t)$ summing $\rho_m(t) \cdot 1_{\{\varphi i \in B_m\}}$ correctly handles the case where a strategy participates in multiple blocks (you get cumulative membership) and correctly zeros out strategies in no admitted block. The product form $p_i = A_i \cdot q_i \cdot M_i \cdot G_i$ preserves the layered structure: admissibility, quality, manifold membership, and PM intent each contribute independently, which is auditable.

Where the spec is thin or worrying.

The parameter defaults are asserted rather than derived. $\tau=4$ with $K=2$ is not innocuous — with two blocks whose Q_m differ by 0.2 in typical operating range, $\rho_{top} \approx 0.69$ and $\rho_{2nd} \approx 0.31$. That's better than hard masking, but you still have ~70% of weight concentrated in one block. With $\tau=6$ the dominant block captures ~77%. The spec needs a parameter-sensitivity sweep that's tied back to V7.3 mixture-template behavior, not just "recommended testing."

There's no sanity check establishing that **V7.4b reduces to V7.3 in some parameter limit**. This should be the first thing demonstrated: pick $(K, \tau, \lambda, \theta_S)$ such that V7.4b reproduces V7.3 default behavior to within Sharpe ±0.02 and DD ±50 bps. If that point exists, V7.4b is a strict generalization and there's no scenario where it should underperform V7.3 *unless its calibration is wrong*. If that point doesn't exist, the soft-aggregation mechanism has a hidden bias that needs to be found before any further tuning.

Block construction is left as three options (connected components, cliques, communities) with no default. This is the most dangerous deferral in the document, because the entire downstream calculation is conditioned on whatever block structure you produce. θ_S sweeps don't help if the clustering algorithm itself is unstable across the threshold range. This needs a recommendation and a stability test (e.g., Jaccard similarity of block partitions across rolling windows).

The **zone collapse problem from V7-Lite is not addressed**. If $Z_t = 1$ still fires ~98% of timesteps, $g(Z_t) = 1.0$ always, and the soft aggregation runs at full gross exposure — which may *amplify* over-concentration through M_i rather than damp it. V7.4b is necessary but not sufficient if zone discipline is still broken. The validation criterion "zone transitions become smoother and structurally interpretable" implicitly acknowledges this but doesn't fix it.

The validation criteria themselves are descriptive rather than operational. "No single block dominates more than 80% of timesteps" — what does "dominate" mean? $\rho_m > 0.5$? > 0.8? Time-fraction of being top-ranked? Each definition produces a different test.

Concrete next steps before sign-off.

The right gate before committing engineering to V7.4b is a four-point check. First, demonstrate the V7.3-recovery point in parameter space. Second, fix block construction to one algorithm with documented stability properties. Third, run a τ sweep at $\{2, 3, 4, 5, 6\} \times K \in \{2, 3\}$ with the AGOR forward-PnL split and chart the (Sharpe, DD, top-block-share) frontier. Fourth, profile zone-state distribution under V7.4b — if it still collapses to Z=1, the soft aggregation is operating in a degenerate regime and the headline numbers will be misleading.

Bottom line: V7.4b is the right *kind* of fix — diagnostically sharp, philosophically aligned, mechanically clean. But it's currently underspecified at exactly the joints that decide whether it works. Treat this document as the design intent, not the implementation contract.

can u suggest next stage?

Next Stage: V7.4c — Operational Hardening Before Dynamic Layers

The architectural diagnosis is now correct. The next stage is **not** more architecture — it's converting V7.4b from a design document into a tested, instrumented, falsifiable engine. Then, and only then, the dynamic layers (V7.5) become meaningful.

Here's the proposed sequence.

Stage 1: V7.4c — Operational Hardening (2–3 weeks)

The four-point gate from the V7.4b evaluation becomes the spec itself.

1.1 Continuity Theorem. Establish that V7.4b is a strict generalization of V7.3, not a parallel architecture. Find the parameter point (K^*, τ^*, λ^*, θ_S^*) where V7.4b reproduces V7.3-Default to within Sharpe ±0.02 and MaxDD ±50bps. This is non-negotiable — if this point doesn't exist, there is a hidden bias in the soft-aggregation that must be located before anything else proceeds. Document it as the *recovery anchor* — every future configuration is then evaluated relative to it.

1.2 Block-Construction Canonicalization. Pick one algorithm. The recommendation: **threshold-graph connected components at $\theta\_S = 0.55$** as the default, with a partition-stability test (Jaccard similarity ≥ 0.7 across adjacent rolling windows). The clique and community options stay in the codebase as research branches, but the production engine has one default. Without this, every downstream result is conditional on a free choice that should not be free.

1.3 Parameter Frontier Mapping. Run the full sweep — $\tau \in \{2, 3, 4, 5, 6\}$, $K \in \{2, 3\}$, $\lambda \in \{0.50, 0.65, 0.80\}$, $\theta\_S \in \{0.45, 0.55, 0.65\}$ — and chart the (Sharpe, MaxDD, top-block-share, turnover) frontier. The objective is not to find the highest Sharpe but to locate the **plateau region** — the contiguous parameter zone where performance is stable. If the frontier is jagged, the engine is overfit to its calibration; if smooth, the architecture is genuine.

1.4 Zone-State Audit. Profile Z_t distribution under V7.4b across the full 2008–2026 sample. If Z=1 still fires ≥95% of timesteps, the soft aggregation is running at full gross exposure and the headline Sharpe is misleading. Zone collapse must be diagnosed independently and fixed *here*, not deferred to V7.5. This was flagged in the V7-Lite brief and remains unresolved.

Deliverable: A V7.4c production engine with locked defaults, a documented parameter plateau, and a zone-state audit that either confirms zone discipline is working or surfaces it as a blocker.

Stage 2: Block-Level Alpha Audit (parallel, 2 weeks)

The V7.3 evaluation document identified the real performance ceiling: "the engine is very good at not dying, increasingly good at steering, but still moderate at extracting convexity. That likely means: block-level alpha engines need strengthening."

This is correct and it's a separate problem from V7.4b. The soft-aggregation fix improves *navigation*; it doesn't improve *exploitation inside admissible regions*. So run an audit:

For each block B_m, compute the realized Sharpe of the within-block allocation (using q_i and current weights) versus equal-weight within the same block over the same timesteps. If within-block alpha is near-zero, no amount of navigation improvement at the manifold layer will translate into materially higher CAGR. This decouples *"is the geometry working?"* (yes — Sharpe and DD prove it) from *"is the within-block exploitation working?"* (open question).

If the audit shows weak within-block alpha, candidate upgrades are: Sharpe-weighted allocation, Kelly-deformed weighting, or a convexity overlay — all of which fit underneath the manifold layer without touching it.

Stage 3: Falsification Suite (1 week)

Before the engine is presented externally, build the tests that would prove the architecture *isn't* working — and run them.

3.1 No-Look-Ahead Audit. Mechanical confirmation that every input at time t depends only on information available at t-1. The V7.3 evaluation made the case philosophically; V7.4c needs the unit tests to back it. Especially: rolling z-scoring windows, volatility estimates, and block construction must all be strictly causal.

3.2 Out-of-Sample Holdout. Reserve 2024–2026 as a true holdout. All V7.4c calibration occurs on 2008–2023. If the Sharpe plateau identified in 1.3 holds in the holdout window, the architecture generalizes. If it collapses, the parameter selection was overfit and Stage 1 needs to repeat with cross-validation.

3.3 Adversarial Regimes. Identify the 5 worst-Sharpe quarters in the sample and run V7.4c against them in isolation. Does the engine survive? Does it survive *better* than V7.3? This is the "did we improve transition stability?" claim from the V7.4b validation criteria, made operational.

Stage 4: Then — V7.5 Dynamic Layers

Only after Stages 1–3 produce a stable, falsifiable, plateau-validated V7.4c does the dynamic layer work make sense. The reason is structural: V7.5 introduces VAIDM, ADD, and Buffett as external modulators. If the underlying engine is still drifting under parameter changes, you cannot tell whether VAIDM is helping or just adding noise that happens to correlate with regime shifts.

V7.5 should then introduce these layers **one at a time**, each gated by the same falsification suite. A useful ordering:

1. ADD first (asset filtering — narrowest scope, easiest to validate)
2. Buffett second (valuation anchor — slow-moving, easy to attribute)
3. VAIDM last (broadest modulator — only meaningful once the others are stable)

Each layer ships with a documented Sharpe/DD delta versus V7.4c-baseline, on the same holdout, with the same plateau test.

Why this sequence over "just add VAIDM next"

The pattern from V7 → V7.3 → V7.4 → V7.4b shows a recurring failure mode: architectural sophistication added on top of insufficiently-validated lower layers. V7.4 broke not because soft aggregation is hard but because the *previous* layer's behavior was implicitly assumed rather than tested. V7.5 will repeat the same mistake if V7.4c isn't operationally hardened first.

The Synergia Conclave takeaway captures this exactly: *"trust cannot be built only by filtering final responses. It must be embedded earlier through simulation, constraint design, steering mechanisms, and mission-level invariants."* The institutional credibility argument from the V7.3 evaluation document — Sharpe 0.75 with DD -8.9%, no hindsight, no leverage tricks — is the asset to protect. Each subsequent version risks that asset unless the falsification gates are in place.

Suggested Deliverables for V7.4c Sign-off

A short PolyAgora V7.4c Validation Note containing:

- The recovery anchor (V7.3-reproduction point in parameter space)
- The canonical block-construction defaults with stability test results
- The parameter frontier chart and identified plateau region
- The zone-state audit (with fix, if zone collapse persists)
- The no-look-ahead audit results
- The OOS holdout performance vs in-sample plateau
- The adversarial-regime performance comparison

This is roughly a 5–8 page document, mostly tables and charts. It's what an institutional allocator would ask for, and it's what makes V7.5 a credible next step rather than another speculative architectural layer.

One-line summary: *V7.4b fixed the architecture; V7.4c proves the architecture; V7.5 extends the architecture. Skipping V7.4c repeats the V7.4 mistake.*